This concludes the proof of Claim 5.5. □

We will next invoke the inductive hypothesis for $E.q$ and w_R. Let us first establish the IH-condition:

IH-condition 5.6. $w_R \in \mathit{to_be_explored}(E.q)$.

Proof of IH-condition 5.6. That $E.q \vdash w_R$ was established in (9). We must prove that $I_{[E.q]}(w_R) \cap \mathit{init_sleep}(E.q) = \emptyset$. By the handling of sleep sets, $\mathit{init_sleep}(E.q)$ may extend $\mathit{init_sleep}(E)$ by some of the processes in $\mathit{done}(E)$. From the assumptions for w we have that for all $p \in \mathit{done}(E)$:

$$\begin{aligned} p \notin WI_{[E]}(w) &\rightarrow p \notin I_{[E]}(w) && \text{by Lemma 4.2(1)} \\ &\rightarrow p \notin I_{[E]}(w_q) && \text{by 5.1.2a} \\ &\rightarrow p \notin I_{[E.q]}(w_q) && \text{by } E \vdash q \diamond w_q\text{, 5.1.3} \end{aligned} \tag{12}$$

Any other process in $\mathit{init_sleep}(E.q)$ was also in $\mathit{init_sleep}(E)$ (passed along by line 10). Again, from the assumptions for w we have that for all $p \in \mathit{init_sleep}(E)$:

$$\begin{aligned} I_{[E]}(w) \cap \mathit{init_sleep}(E) = \emptyset &\rightarrow p \notin I_{[E]}(w) \\ &\rightarrow p \notin I_{[E.q]}(w_q) && \text{(same reasoning)} \end{aligned} \tag{13}$$

Since in (12) and (13) we considered $I_{[E.q]}(w_q)$ the only process that can be in $I_{[E.q]}(w_R) \cap \mathit{init_sleep}(E.q)$ is $\widehat{e_q}$. We have two cases.

- If there is no event e in w_q such that $e \rightarrow_{E.w_R} e_q$, then by the construction of e_q, we have $E \nvdash q \diamond \widehat{e_q}$, implying $\widehat{e_q} \notin \mathit{init_sleep}(E.q)$ (as it is filtered by line 10).
- Otherwise, let e be the last event in w_q such that $e \rightarrow_{E.w_R} e_q$. Let w'_q be the prefix of w_q that precedes e, and let w''_q be the suffix of w_q that follows e, i.e., $w_q = w'_q.\widehat{e}.w''_q$. By construction, we have $E.w'_q \nvdash \widehat{e} \diamond \widehat{e_q}$ and $E.w'_q \vdash \widehat{e} \diamond q$. By Property (7) of Definition 3.2, we infer $E.w'_q \nvdash \widehat{e} \diamond (q.\widehat{e_q})$, which implies $e \rightarrow_{E.w_q.q.\widehat{e_q}} e_q$, which implies $e \rightarrow_{E.q.w_q.\widehat{e_q}} e_q$. Hence $e \rightarrow_{E.q.w_R} e_q$, and so $\widehat{e_q} \notin I_{[E.q]}(w_R)$.

This concludes the establishment of IH-condition 5.6. □

Using the inductive hypothesis for $E.q$, the algorithm explores some sequence of the form $E.q.w'$ such that:

$$E.q.w' \simeq E.q.w_R.v' \text{ for some } v' \tag{14}$$

Note that in w', the event e_q exists, but need not be the last occurring of the events e_p for $p \in \mathit{done}(E)$, since such independent events may have been reordered from $w_R.v'$ to w'.

Let w'_q be the prefix of w' up to, but not including e_q. By the construction of w_q, we have $\mathit{next}_{[E]}(q) \precsim_{E.q.w_q} e_q$. From (14), it also follows that $\mathit{next}_{[E]}(q) \precsim_{E.q.w'_q} e_q$ (this is the same race as the race $\mathit{next}_{[E]}(q) \precsim_{E.q.w_q} e_q$, i.e., between $\mathit{next}_{[E]}(q)$ and e_q over the sequence w_q). Since the sequence $E.q.w'$ is one actually explored by the algorithm, line 5 will detect the race $\mathit{next}_{[E]}(q) \precsim_{E.q.w'_q} e_q$. At that point,

- p in the algorithm will correspond to $\widehat{e_q}$,
- E' in the algorithm will correspond to E, and
- e in the algorithm will correspond to $\mathit{next}_{[E]}(q)$.

The algorithm will extract (lines 6 and 7) the sequence $v = \mathit{notdep}(\mathit{next}_{[E]}(q), E.q.w'_q).\widehat{e_q}$ from $E.q.w'_q.\widehat{e_q}$ by removing events that have $\mathit{next}_{[E]}(q)$ happening-before them and identify its initials $I_{[E]}(v)$. Below let us use $\xrightarrow{*}$ to denote multiple steps. For any such $q' \in I_{[E]}(v)$ it follows that: